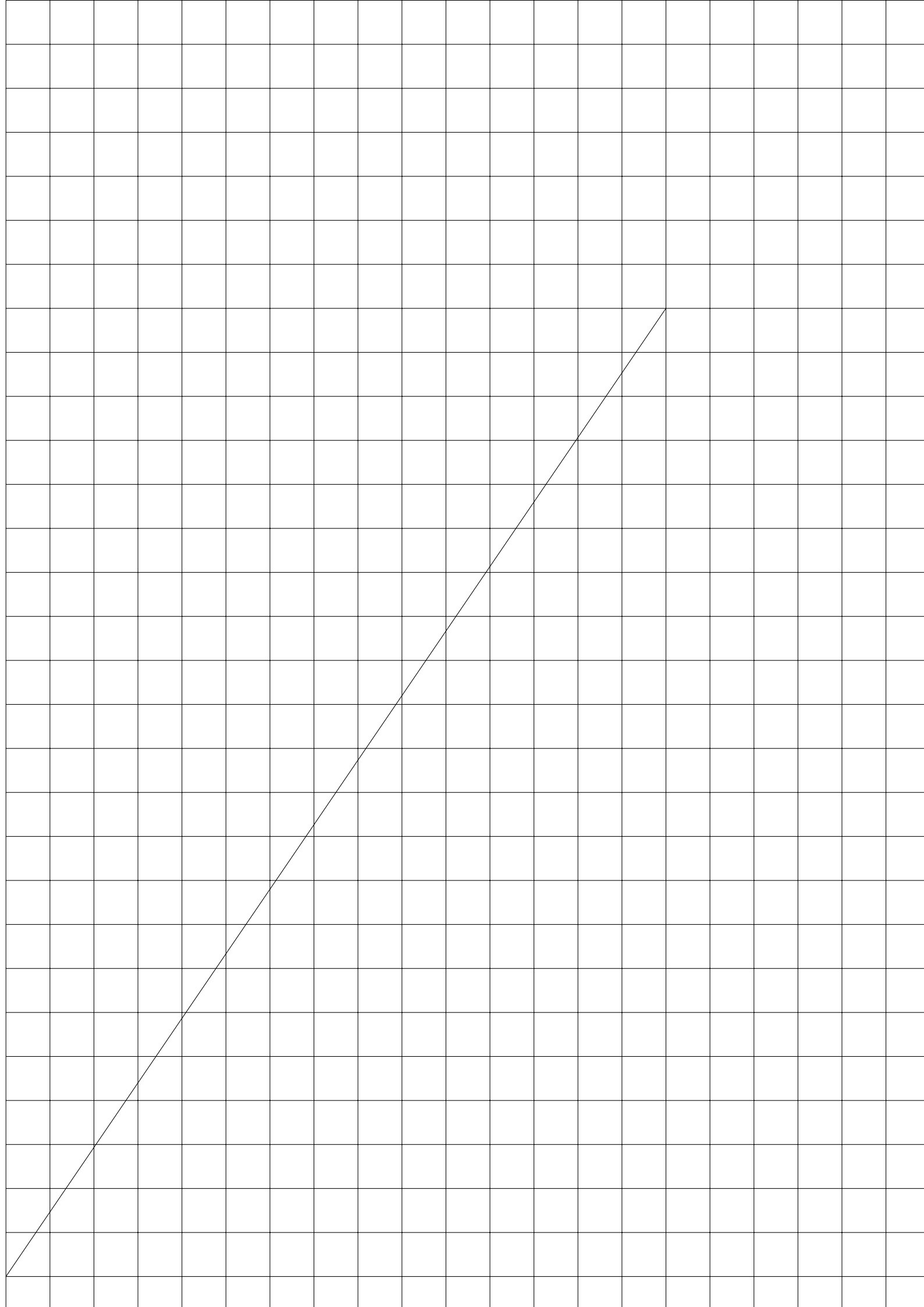




$\text{\LaTeX}$ Template

Marco Ramos¹

¹Coventry University, Coventry, UK

March 15, 2025



Made with $\text{\LaTeX}$

_303COM Declaration of originality

I Declare that This project is all my own work and has not been copied in part or in whole from any other source except where duly acknowledged. As such, all use of previously published work (from books, journals, magazines, internet etc.) has been acknowledged by citation within the main report to an item in the References or Bibliography lists. I also agree that an electronic copy of this project may be stored and used for the purposes of plagiarism prevention and detection.

_Statement of copyright

I acknowledge that the copyright of this project report, and any product developed as part of the project, belong to Coventry University. Support, including funding, is available to commercialize products and services developed by staff and students. Any revenue that is generated is split with the inventor/s of the product or service. For further information, please see www.coventry.ac.uk/ipr or contact ipr@coventry.ac.uk.

_Statement of copyright

I declare that a proposal for this project has been submitted to the Coventry University ethics monitoring website (<https://ethics.coventry.ac.uk/>) and that the application number is listed below (Note: Projects without an ethical application number will be rejected for marking).

Signed: Date: March 15, 2025

Please complete all fields.

First Name	Marco
Last Name	Ramos
Student ID Number	10415201
Ethics Application Number	P183438
1 st Supervisor Name	Dr Dianabasi Nkantah
2 nd Supervisor Name	

This form must be completed, scanned, and included with your project submission to Turnitin. Failure to append these declarations may result in your project being rejected for marking



_Abstract

Abstract inserted here.



_Contents

1	Introduction	10
1.1	Background & Context	10
1.2	Motivation & Rationale	10
1.3	Research Question & Objectives	10
1.4	Scope & Limitations	10
1.5	Overview of Report Structure	10
1.6	Assessment Against Objectives	11
1.7	Strengths, Weaknesses & Limitations	11
1.8	Discussion of Unforeseen Challenges & Adaptations	11
1.9	Recommendations for Future Work	11
2	Literature Review	12
2.1	Overview of Existing Technologies & Theoretical Foundations	12
2.2	Technical Review	12
2.3	Research Review	12
2.4	Identified Gaps & Opportunities	12
2.5	Proposed Innovation & Conceptual Framework	13
3	Project Management & Planning	14
3.1	Project Timeline & Milestones	14
3.2	Task Breakdown & Resource Allocation	14
3.3	Risk Management & Mitigation Strategies	14
4	Research Methodology	15
4.1	Research Design	15
4.2	Data Collection & Analysis Methods	15
4.3	Ethical Considerations & Approvals	15
4.4	Methodological Limitations	15
5	Technical Aspects & System Design	16
5.1	System Architecture and Design	16
5.2	Technology Stack & Development Environment	16
5.3	Data Structures, Algorithms & Machine Learning Components	16
5.4	Integration & Interoperability	16
6	Implementation & Development Process	17
6.1	Prototyping & Iterative Development	17
6.2	Version Control, Documentation & Quality Assurance	17
6.3	Testing & Validation	17
7	Results & Analysis	18
7.1	Presentation of Results	18
7.2	Comparative Analysis	18



7.3 Data Interpretation & Discussion	18
8 Conclusion & Reflections	19
8.1 Summary of Findings & Outcomes	19
8.2 Conclusions	19
8.2.1 Addressing the Research Question	19
8.3 Personal Reflection & Learning	19
8.3.1 Reflection on the Project Management Process	19
8.3.2 Reflections on Technical & Methodological Choices	19
8.3.3 Professional Growth & Future Directions	19
8.4 Final Recommendations & Future Work	19



_List of Figures



_List of Tables



_Source Code



Chapter 1

INTRODUCTION

1.1. _Background & Context

- Historical overview of the problem(e.g. disc management challenges)
- Relevance to current technological and user needs

1.2. _Motivation & Rationale

- Why the project matters
- Identified shortcomings in current solutions

1.3. _Research Question & Objectives

- Research question
- Project goals

1.4. _Scope & Limitations

- What is included (and excluded) in the project

1.5. _Overview of Report Structure

- Brief roadmap of the chapters and content



1.6. _Assessment Against Objectives

- How well the project meets the stated aims and research question

1.7. _Strengths, Weaknesses & Limitations

- A critique of what worked and what did not

1.8. _Discussion of Unforeseen Challenges & Adaptations

- Reflection on project changes, lessons learned, and improvement areas

1.9. _Recommendations for Future Work

- Suggestions for further research or system enhancements



Chapter 2

LITERATURE REVIEW

2.1. _Overview of Existing Technologies & Theoretical Foundations

- Summary of current disc management and automation tools
- Key concepts (file systems, directory automation, machine learning for classification)

2.2. _Technical Review

- Analysis of leading disc management tools (e.g. WinDirStat, WizTree, TreeSize)
- Discussion of system performance, interface design, and technical limitations

2.3. _Research Review

- Operating System Fundamentals (file systems, Master File Table, etc.)
- Theoretical methods for automation and categorization
- Review of related academic and industry studies

2.4. _Identified Gaps & Opportunities

- What existing tools lack
- Opportunities for innovation in this project



2.5. _Proposed Innovation & Conceptual Framework

- How this project aims to fill the identified gaps
- Summary of innovative features (e.g. smart file categorization, machine learning integration)



Chapter 3

PROJECT MANAGEMENT & PLANNING

3.1. _Project Timeline & Milestones

- Timeline/planning methodology

3.2. _Task Breakdown & Resource Allocation

- Specific work packages, deadlines, and responsibilities

3.3. _Risk Management & Mitigation Strategies

- Identifying potential risks and contingency plans



Chapter 4

RESEARCH METHODOLOGY

4.1. _Research Design

- Explanation of the overall approach (experimental, case study, etc.)

4.2. _Data Collection & Analysis Methods

- Techniques (e.g. user testing)

4.3. _Ethical Considerations & Approvals

- Discussion of ethics, informed consent, and application details

4.4. _Methodological Limitations

- Acknowledgement of any constraints or potential biases



Chapter 5

TECHNICAL ASPECTS & SYSTEM DESIGN

5.1. _System Architecture and Design

- Overall blueprint of the software/hardware solution

5.2. _Technology Stack & Development Environment

- Programming languages, frameworks, libraries, and hardware requirements - Thought process for technology choices

5.3. _Data Structures, Algorithms & Machine Learning Components

- Detailed description of core technical components and innovations

5.4. _Integration & Interoperability

- How the system components interact and integrate with existing tools



Chapter 6

IMPLEMENTATION & DEVELOPMENT PROCESS

6.1. _Prototyping & Iterative Development

- Description of development stages and prototype iterations

6.2. _Version Control, Documentation & Quality Assurance

- Tools and processes for ensuring code quality and traceability

6.3. _Testing & Validation

- (e.g. Unit, integration, and performance testing methods) - Test cases and results overview



Chapter 7

RESULTS & ANALYSIS

7.1. _Presentation of Results

- Quantitative findings (metrics, benchmarks) - Qualitative observations (user feedback, case study insights)

7.2. _Comparative Analysis

- Evaluation against existing solutions and initial objectives

7.3. _Data Interpretation & Discussion

- In-depth analysis of results in the context of your research question



Chapter 8

CONCLUSION & REFLECTIONS

8.1. _Summary of Findings & Outcomes

- Recap of key achievements and discoveries

8.2. _Conclusions

8.2.1. _Addressing the Research Question

- How the project answered the research question

8.3. _Personal Reflection & Learning

8.3.1. _Reflection on the Project Management Process

- What worked well, challenges faced, and supervision insights

8.3.2. _Reflections on Technical & Methodological Choices

- Self-assessment of the development and research methods

8.3.3. _Professional Growth & Future Directions

- Lessons learned and advice for future projects or successors

8.4. _Final Recommendations & Future Work

- Proposed next steps and potential improvements for further research



_References

Hoffman, C., & Lewis, N. (2017, June 21). The 5 Best Free Tools to Analyze Hard Drive Space on Windows. How-To Geek. Retrieved December 9, 2024, from <https://www.howtogeek.com/113012/10-best-free-tools-to-analyze-hard-drive-space-on-your-windows-pc/>

SpaceSniffer features. (n.d.). Retrieved December 9, 2024, from http://www.uderzo.it/main_products/space_sniffer/features.html

